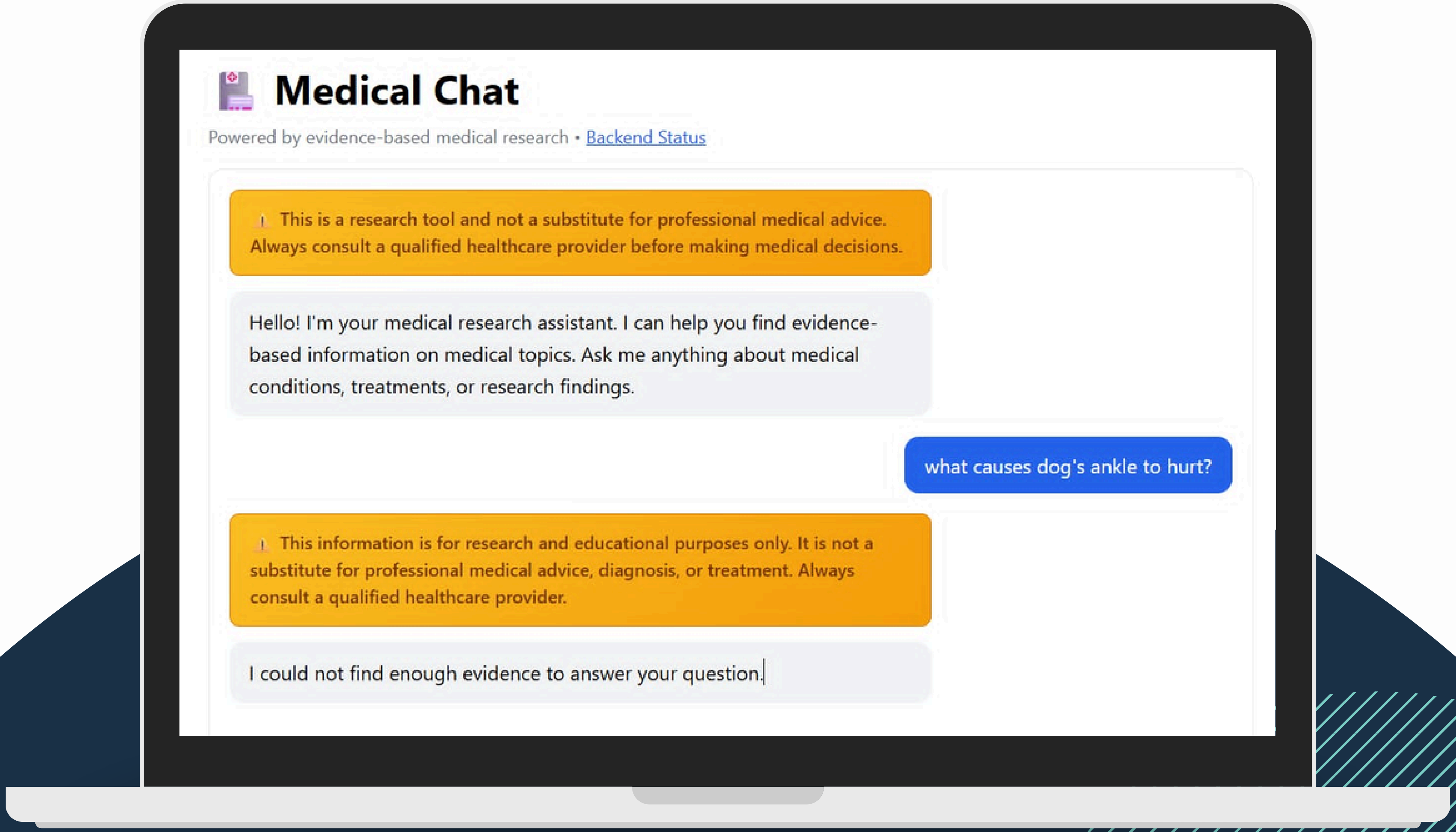
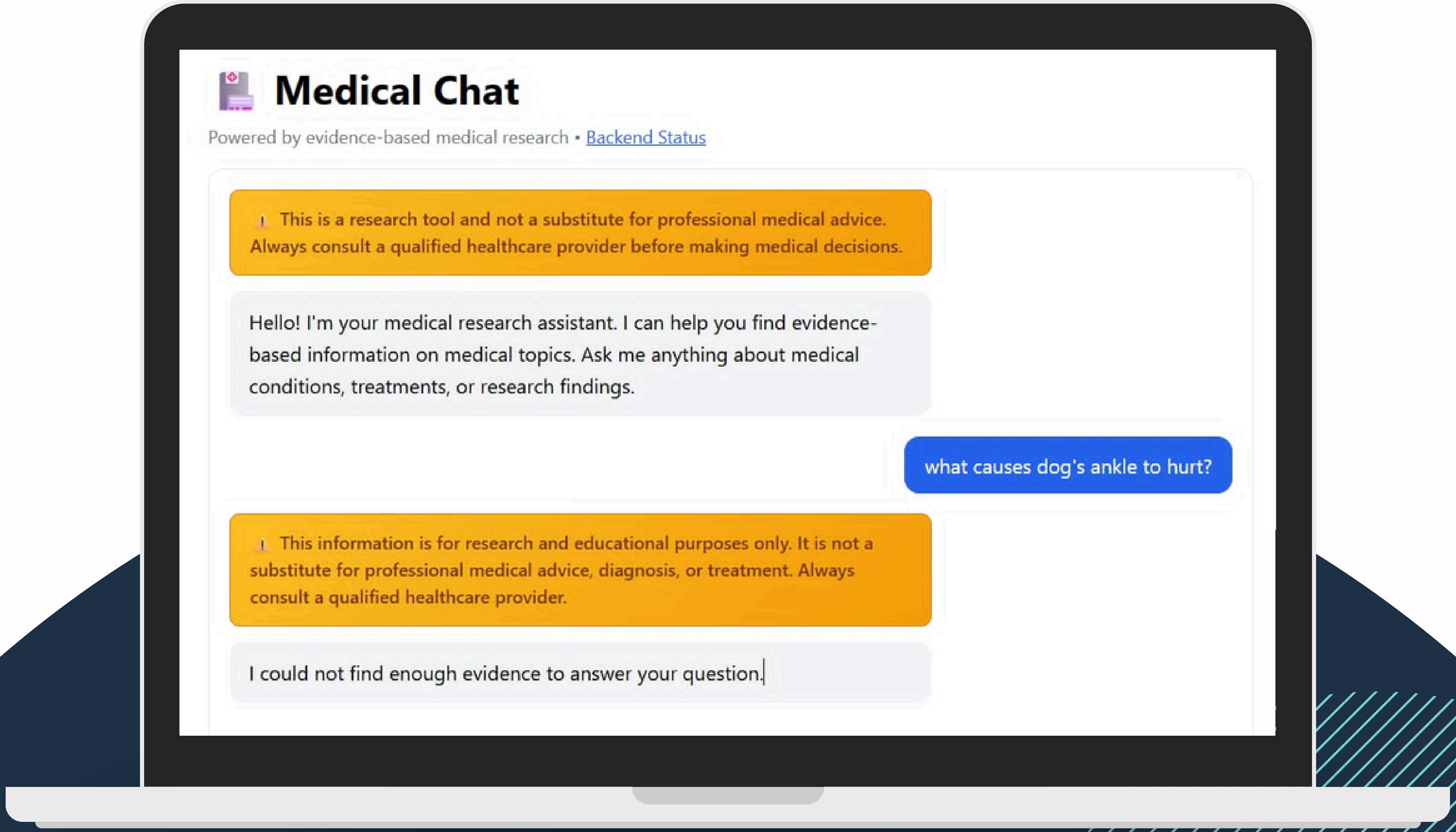
Medical Chat
AI-Powered Medical Assistant



Github

https://github.com/Antonina
Gardzielewska/medical-chat



Project overwiev

Medical Chat is an AI-powered medical chat assistant designed to provide evidence-based informational responses using:

- FastAPI (Python) backend
- Next.js (Node.js) frontend
- RAG (Retrieval-Augmented Generation) pipeline
- ChromaDB for vector search
- Ollama for local LLM inference

github link:
https://github.com/AntoninaGardzielewska/medical-chat

Key Idea

Build a conversational assistant that can answer medical questions using retrieved evidence from PubMed articles.

Important disclaimers: This project is an experimental proof of concept and does not provide medical advice.

⚠️ This information is for research and educational purposes only. It is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider.

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Architecture

- **Frontend**
  - Built with Next.js
  - Provides the chat interface for users
  - Communicates with the backend through REST API calls
- **Backend**
  - Built with FastAPI
  - Exposes the /api/v1/chat endpoint
  - Implements the RAG pipeline:
    - Query rewriting
    - Vector retrieval
    - Answer synthesis with Ollama
- **Vector Database**
  - ChromaDB stores embeddings of medical article chunks
  - Enables semantic similarity search
- **LLM Layer**
  - Ollama runs the language model locally
  - Current default model: llama3.2:latest
- **Data Source**
  - PubMed medical articles
  - Retrieved via the NCBI E-utilities API

CI/CD

GitHub Actions runs on every push and pull request to main:

- Tests — pytest with coverage (Python 3.12 & 3.13)
- Style — Ruff lint + format check
- Types — mypy strict mode
- Security — Bandit SAST scan
- Coverage — uploaded to Codecov

Data Ingestion Pipeline

1. Fetch PubMed articles.
2. Chunk article text into smaller passages.
3. Generate embeddings for each chunk.
4. Store vectors and metadata in ChromaDB.

Key implementation details

- Implemented in backend/ingestion/run_ingestion.py.
- Creates the local database at backend/chroma_db.
- Populates the pubmed_abstracts collection.

How We Build the knowledge base

Step 1 - Fetch PubMed articles

Implemented in backend/ingestion/fetch.py.

Selection criteria

We query PubMed using predefined medical keywords: (because of our hardware limitations)

- “type 2 diabetes”
- “hypertension”
- “heart failure”

API parameters

- retmax=1000 → fetch up to 1000 results per term
- sort=relevance → PubMed returns the most relevant articles first
- Articles are retrieved in batches of 200 IDs

Filtering

We keep articles only if they contain:

- PMID
- Publication year
- Title
- Abstract text

Result

The script stores the selected articles in articles.json.

How We Build the knowledge base

Step 2 – Chunk the articles
Implemented in backend/ingestion/chunk.py.

Large articles are split into smaller semantic pieces so retrieval is more precise.
Chunking configuration
- Chunk size: 256 tokens
- Chunk overlap: 32 tokens
- Tokenizer: PubMedBERT embeddings tokenizer

Output
Each chunk contains:
- Text fragment
- Article metadata (title, authors, journal, year, PMID)
- Unique chunk ID

The chunks are saved to chunked_articles.json.

How We Build the knowledge base

```
============================================================
  1. TOTAL CHUNKS IN DATABASE
============================================================
  Total chunks stored: 7,034

============================================================
  2. RANDOM SAMPLE (5 chunks)
============================================================

  --- Chunk 1 ---
  PMID    : 34251351
  Title   : Type 2 Diabetes Mellitus and Cognitive Impairment.
  Journal : Acta medica Indonesiana
  Year    : 2021
  Text    : Type 2 diabetes mellitus (T2DM) is strongly associated with lower performance on multiple domains of cognitive function and with structural abnormalities of t
he brain. With the growing epidemic of dia...

  --- Chunk 2 ---
  PMID    : 35919809
  Title   : Empagliflozin in the treatment of heart failure and type 2 diabetes mellitus: Evidence from several large clinical trials.
  Journal : International journal of medical sciences
  Year    : 2022
  Text    : Heart failure coexists with type 2 diabetes mellitus, which seriously affects the clinical treatment and prognosis. At present, the treatment for patients wit
h established heart failure and type 2 dia...

  --- Chunk 3 ---
  PMID    : 30193140
  Title   : Schizophrenia in type 2 diabetes mellitus: Prevalence and clinical characteristics.
  Journal : European psychiatry : the journal of the Association of European Psychiatrists
  Year    : 2019
  Text    : This study investigated the prevalence and characteristics of schizophrenia in patients with type 2 diabetes mellitus (T2DM) in Taiwan.

  --- Chunk 4 ---
  PMID    : 28322073
  Title   : SGLT2 inhibitor/DPP-4 inhibitor combination therapy - complementary mechanisms of action for management of type 2 diabetes mellitus.
  Journal : Postgraduate medicine
  Year    : 2017
  Text    : empagliflozin and linagliptin or dapagliflozin and saxagliptin in the management of type 2 diabetes mellitus. Both combinations have been approved as single-p
ill formulations.

  --- Chunk 5 ---
  PMID    : 35056324
  Title   : Impaired Folate-Mediated One-Carbon Metabolism in Type 2 Diabetes, Late-Onset Alzheimer's Disease and Long COVID.
  Journal : Medicina (Kaunas, Lithuania)
  Year    : 2022
  Text    : Impaired folate-mediated one-carbon metabolism (FOCM) is associated with many pathologies and developmental abnormalities. FOCM is a metabolic network of inte
rdependent biosynthetic pathways that is k...
```

How We Build the knowledge base

Step 3 – Embed and store

Implemented in backend/ingestion/embed_and_store.py.

Embedding model

- PubMedBERT-based sentence embeddings
- Optimized for biomedical text

Storage

- Embeddings are inserted into ChromaDB
- Collection name: pubmed_abstracts
- Persistence path:
    - Local dev: backend/chroma_db
    - Docker: /data/chroma_db
- number of articles: 2156

Runtime Query Pipeline (RAG)

What happens when a user asks a question?

Step 1 – User query

Example:

“What are the symptoms of heart failure?”

Step 2 – Retrieve relevant chunks

Implemented in backend/rag/retrieve.py.

Retrieval process

- The query is embedded into a vector.
- ChromaDB searches for semantically similar chunks.
- Results are filtered by similarity distance:
  - Only chunks with distance < max_distance are kept
  - The number of returned chunks is capped at max_results

Step 3 – Synthesize the answer

Implemented in backend/rag/synthesize.py and backend/rag/llm.py.

LLM workflow

- The retrieved chunks are combined into a prompt.
- The prompt is sent to Ollama via HTTP:
  - POST /api/generate
- Ollama generates a final natural-language answer grounded in the retrieved evidence.

Step 4 – Return response

The backend returns:

- generated answer
- optionally, metadata about supporting articles

How We Build the knowledge base

```
============================================================
  3. SEMANTIC SEARCH TEST
============================================================
  Query: 'treatment options for type 2 diabetes'

  Result 1  (distance: 95.7119)
  ID       : 18514088_0
  PMID     : 18514088
  Title    : Intensifying treatment in poorly controlled type 2 diabetes mellitus: case reports.
  Text     : There is no single correct treatment option for all patients with type 2 diabetes mellitus. Duration of diabetes and stage of the disease, level of control, l
ifestyle habits, and attitude toward disea...

  Result 2  (distance: 107.6119)
  ID       : 36650625_0
  PMID     : 36650625
  Title    : Evolution of Guideline Recommendations on Insulin Therapy in Type 2 Diabetes Mellitus Over the Last Two Decades: A Narrative Review.
  Text     : The prevalence of type 2 diabetes mellitus has been increasing worldwide. As the therapeutic options for type 2 diabetes mellitus have evolved over the last 2
 decades, national and global guidelines r...

  Result 3  (distance: 109.8022)
  ID       : 35259045_0
  PMID     : 35259045
  Title    : Cost-effectiveness of dapagliflozin compared to DPP-4 inhibitors as combination therapy with metformin in the treatment of type 2 diabetes mellitus without es
tablished cardiovascular disease in Colombia.
  Text     : SGLT2 inhibitors or DPP-4 inhibitors are among the preferred options in patients with type 2 diabetes mellitus (T2DM) without established cardiovascular disea
se.

  Result 4  (distance: 109.8634)
  ID       : 29124701_0
  PMID     : 29124701
  Title    : Vitamin D and Type 2 Diabetes Mellitus.
  Text     : Type 2 diabetes mellitus (T2DM) has become a significant global health care problem and its reported incidence is increasing at an alarming rate. Despite the
improvement in therapy and development of ...

  Result 5  (distance: 111.7144)
  ID       : 21718606_0
  PMID     : 21718606
  Title    : Comprehensive treatment for type 2 diabetes mellitus.
  Text     : The treatment of type 2 diabetes mellitus (T2DM) is traditionally focused on glycemic control. In recent years, comprehensive treatment approaches including b
lood sugar control, blood pressure managem...
```

Example end-to-end flow

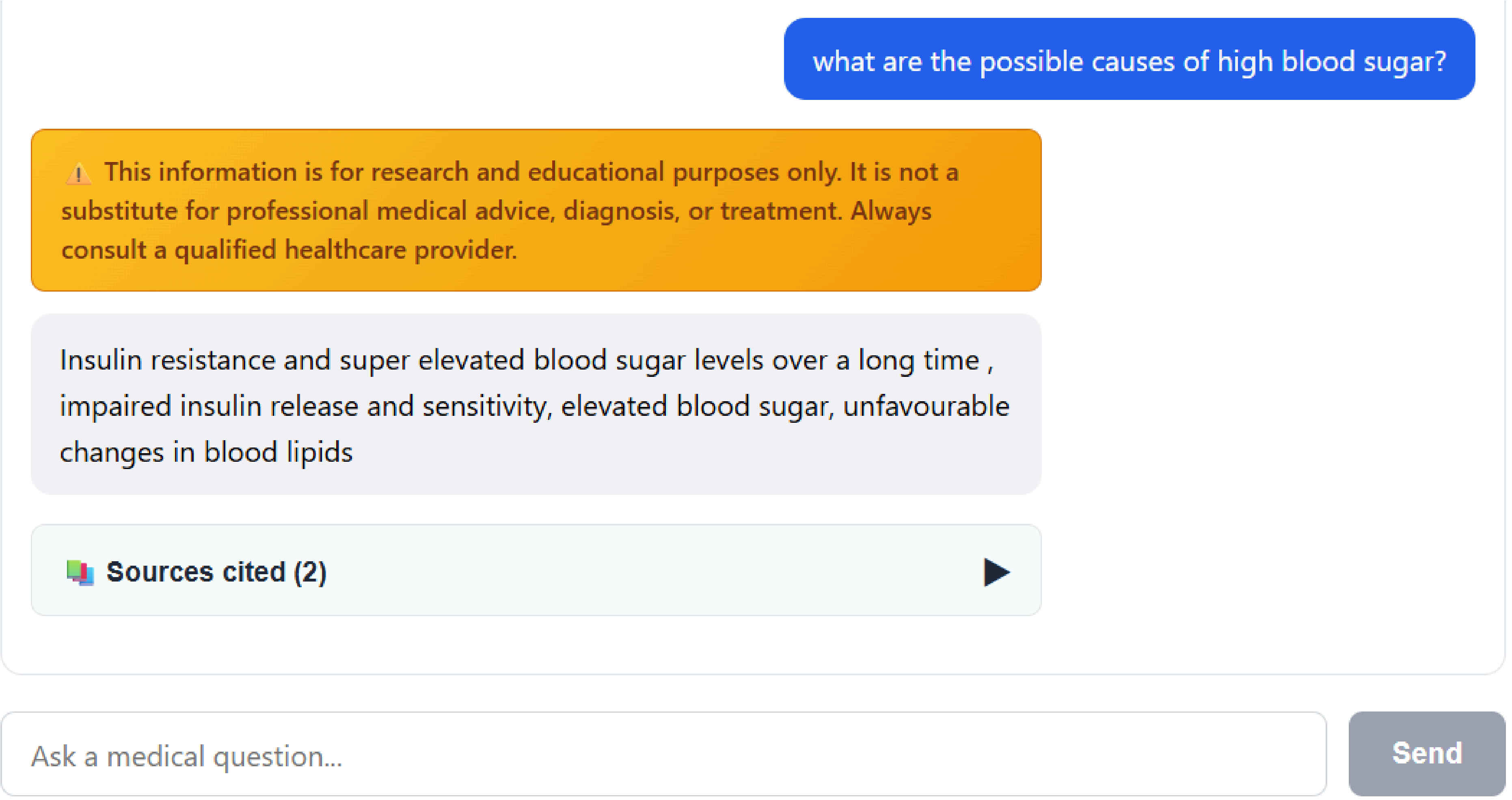
Example query

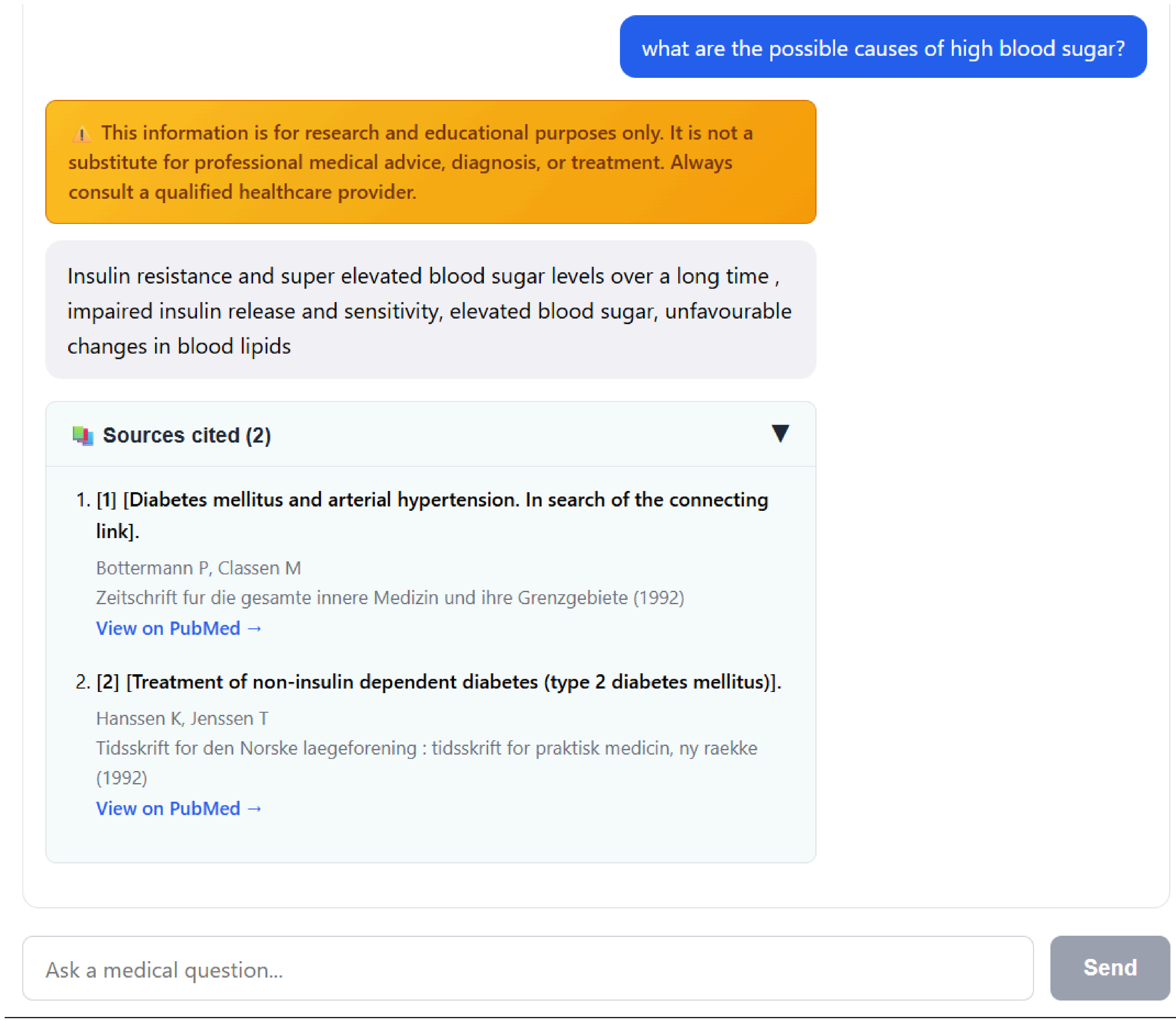
“What are common treatments for type 2 diabetes?”

System flow

- User sends question from the Next.js frontend.
- FastAPI receives the request at /api/v1/chat.
- Retriever searches ChromaDB for diabetes-related chunks.
- Top relevant chunks are selected and filtered by similarity distance.
- Ollama synthesizes an evidence-based answer using the retrieved context.
- Frontend displays the final response to the user.

Example answers





Example answers

Medical Chat

Powered by evidence-based medical research • Backend Status

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Hello! I'm your medical research assistant. I can help you find evidence-based information on medical topics. Ask me anything about medical conditions, treatments, or research findings.

what causes dog's ankle to hurt?

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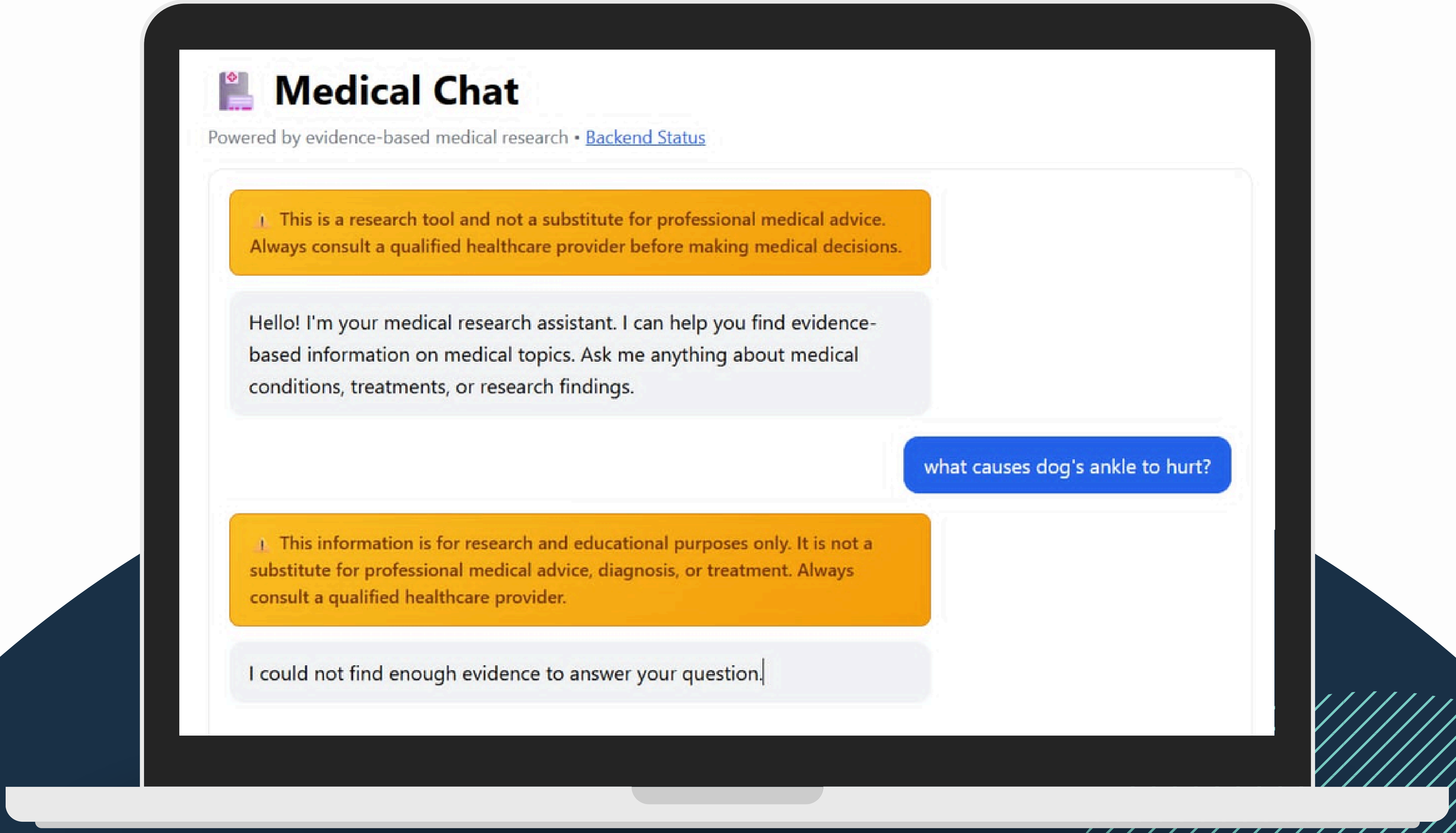
I could not find enough evidence to answer your question.

Ask a medical question...

Send

Thank you!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